

21/09/2026

Recaping to date, I have:

- Set up labbook
- Set up code repository locally
- Received code and basic introduction from Dr Turci
- Partly read introductory papers/books on the topic

Looking forwards, there are 2 more immediate things i need to do:

- continue reading to build up understanding:
 - Second quantisation (Altland and Simons 2010)
 - Introductory paper (Rivas and Huelga 2012)
- Build up main “computational engine”:
 - Look through and understand code Dr Turci has sent me
 - Refactor said code into a convenient utility I can use throughout this project

After i’ve done these i’ll be equipped with the main theoretical and computational tools for this project. This will inform the rest of the investigation of this project.

Having a practical understanding of what this project entails should help me understand the theory “in context”. As such, i think for now its worth examining the code Dr Turci has sent me, in order to start planning the computational engine.

Examining the provided julia code (Y4DissertationCodeMaterials/providedCode/basic-1site-bath-julia.ipynb), it starts by reiterating the maths. Note; explanation recommends finding a good explanation for “wick’s theorem”. Examining my old personal notes for QFT, i recall it being relevant to evaluating the non commuting terms in the Dyson series. That understanding was based on (Schwichtenberg 2020) which may be of use. However i wonder if the second quantisation (Altland and Simons 2010) chapter will introduce it in a more contextually relevant manner. Skimming the chapter it seems so... Changed my mind, i’m going to try and quickly skim the chapter with

the remainder of this afternoon. I expect the concepts will be familiar, so shouldn't take too long. Code examination can come tmr.

Reading notes: Second quantisation impressions (Altland and Simons 2010):

Second quantisation seems to be general framework of ladder operators and associated commutation relations; which is a familiar concept. Apparently generalises nicely to N-body quantum physics.

Chapter motivates second quantisation by highlighting computational issues w/many body QM due to indistinguishability, symmetrization, antisymmetrization, and the exchange of particle numbers.

Second quantisation basics

It then covers the “apparatus of second quantization”.

Compares representation of labelling particles in state individually to simply counting the number of particles in each state (I.E $|1, 1, 1, 1, 2, 2, 3\rangle$ becomes $|4, 2, 1\rangle$). This is the “occupation number representation”. Also note the “Fock space” $\mathcal{F}$; made up of basis states $\{|n_i\rangle\}$ w/out occupation num condition $\sum n_i = N$. Therefore many body state can be represented as $|\Psi\rangle = \sum_{n_i} c_{n_i} |n_i\rangle$.

Within this framework, ladder operators can be constructed to act on $\mathcal{F}$:

$$a_i^\dagger |n_1, \dots, n_i, \dots\rangle = (n_i + 1)^{1/2} \zeta^{s_i} |n_1, \dots, n_i + 1, \dots\rangle \quad s_i = \sum_{j=1}^{i-1} n_j$$

Note that in the fermionic case, the $n_i + 1$ operation is a mod 2 operation $n_i + 1 \equiv (n_i + 1) \% 2$ Utility of this; can generate all states by repeated application of $a_i^\dagger$ to vacuum state:

$$|n_1, \dots\rangle = \prod_i \frac{1}{(n_i!)^{1/2}} (a_i^\dagger)^{n_i} |0\rangle$$

Note that:

Wierd notation used by chapter for commutation/anti-commutation relations:

$$[A, B]_\zeta = AB - \zeta BA$$

$$[A, B]_{\zeta=+1} = [A, B]_+ = AB - BA \quad // \text{commutator}$$

$$[A, B]_{\zeta=-1} = [A, B]_- = AB + BA = \{A, B\} \quad // \text{anticommutator}$$

$$[a_i, a_j^\dagger]_\zeta = \delta_{ij}$$

$$[a_i, a_j]_\zeta = 0$$

$$[a_i^\dagger, a_j^\dagger]_\zeta = 0$$

practical aspects:

The chapter continues into the “practical aspects” of this framework.

i Single particle base transforms

Firstly; switching between single particle bases $\{\lambda\} \rightarrow \{\tilde{\lambda}\}$
- the operators transform as:

$$a_{\tilde{\lambda}}^\dagger = \sum_{\lambda} \langle \lambda | \tilde{\lambda} \rangle a_{\lambda}^\dagger$$

$$a_{\tilde{\lambda}} = \sum_{\lambda} \langle \tilde{\lambda} | \lambda \rangle a_{\lambda}$$

where λ represents a particular particle state in a particular single particle basis. For continuous quantum numbers (I.E, pos) the sum becomes integrals as:

$$a_{\lambda} = \int_0^L dx \langle \lambda | x \rangle a(x)$$

$$a(x) = \sum_k \langle x | \lambda \rangle a_{\lambda}$$

i Single particle operators

Secondly operators acting on single particles w/single party operator o_n on nth particle typically become $\mathcal{O}_1 = \sum_{n=1}^N o_n$.

Typical example noted is momentum: $T = \sum_n \frac{p_n^2}{2m}$

Also introduces “occupation number operator” which counts #particles in state:

$$n_{\lambda} = a_{\lambda}^\dagger a_{\lambda}$$

Which is useful for actually constructing $\mathcal{O}_1$. Given $o = \sum_i o_{\lambda_i} |\lambda_i\rangle \langle \lambda_i|$, $o_{\lambda_i} = \langle \lambda_i | o | \lambda_i \rangle$. Skipping over details i’m uninterested in rn, the general result is:

$$\mathcal{O}_1 = \sum_{\mu\nu} \langle \mu | o | \nu \rangle a_{\lambda}^\dagger a_{\lambda}$$

The chapter then goes into concrete examples to cement these ideas - not immediately relevant, but earmarking p48 in case useful for later.

skimming rest of chapter

There's also a practical applications section. Not going to go into that in depth; skimming to try to earmark anything of use:

- Electrons in a periodic potential (goes through nearly free electrons -> tight binding)(p51)
- Mott-hubbard transition and the magnetic state (p61)
- interacting fermions in one dimension (p67)
 - this section starts using ideas of interaction vs non interacting terms - i suspect wicks theorem (and other useful bits) may be here
- Quantum spin chains (76)
- Summary and outlook - Ch summary (p83)

Wick theorem is not in this section - its mentioned in the summary. ctrl+f through the book doesn't seem to have a contained explanation albeit many uses of Wick's theorem. In which case (Schwichtenberg 2020) may be a better call - the mathematical machinery seems to generalize beyond QFT, so it should be applicable. Ctrl+f through (Schwichtenberg 2020), the explanation of contractions/Wick's theorem goes between p426-p428 at minimum (more after that). Not a today thing to look at but worth earmarking for later.

This reading was a useful refresher and "earmarking exploration" - i think tmr i can go through the code in depth to understand what its doing/how its structured. I should be able to come up with a near term plan thereafter (wether i can get on w/the computational engine or if i need more reading).

22/09/2026

Going to look through the code Dr Turci sent me. I think task list is:

- Read through code and break down what its doing
- Compare the julia vs python code - quick runtime benchmark should show Julia to be faster, thus justifying its use

- think about how to rewrite the simulation into a utility that can be used for the rest of the project:
 - Assuming Julia's faster, write the simulation code in Julia
 - If needbe, can export the simulation results, and build a python julia handler - that way the julia sim is smth that can be called from python code as its own utility

Examining provided code:

First code block is just some library setup.

Second code block sets up spectral function (currently set to a semi-circle) as function of ω . It then initialises an empty 2D hamiltonian array for the bath sites. Unsure why 2D; my understanding is that a 1D list would encapsulate the bath perfectly fine. Potentially initialising all timesteps and fills them throughout the sim? Ooooooh, nvm, its the hamiltonian matrix. Plots the spectral function over ω .

Third code block sets up initial state using Fermi Dirac stats:

$$f(\beta, \mu, E) = \frac{1}{1 + e^{\beta E - \beta \mu}}$$

The simulation assumes no initial correlations; therefore initial correlation matrix is diagonalised and filled with $f(\beta, \mu, E)$ for all corresponding system and bath sites:

$$C_{ii} = f(\beta, \mu, E_i)$$

Third code block sets up this initial correlation matrix.

Fourth/fifth code blocks illustrate fact that $C_{t=0}$ is diagonalized in site, not hamiltonian basis.

Sixth/seventh code block builds unitary time evolution:

$$U(t) = e^{-iHt}$$

And build function to evolve correlation matrix over time t. :::
{.column-margin} Sidenote; another julia syntax note; matrix complex conjugate is indicated as H' :::

Sidenote; just realized Julia indicates vector maths using a "." notation (I.E, scalar addition $\mathbf{a}+\mathbf{b}$ becomes vector addition $\mathbf{a}.\mathbf{+b}$)

Code block 9 (and surrounding commented out code blocks) compute the correlation matrix over a range of times. The commented out bits inc a computationally slower algorithm (the main method is iterative), and a commented out plot.

Code blocks 10, 11 compute and plot the reduced density matrix element corresponding to the occupation of the system. Recalling $\rho @$

Altland, Alexander, and Ben Simons. 2010. *Condensed Matter Field Theory 2e*. Cambridge University Press. <https://doi.org/https://doi.org/10.1017/CBO9780511789984>.

Rivas, Ángel, and Susana F Huelga. 2012. *Open Quantum Systems. An Introduction*. <https://doi.org/https://doi.org/10.48550/arXiv.1104.5242>.

Schwichtenberg. 2020. *No-Nonsense Quantum Field Theory*.